a

Energy

Number frequency

1.0
0.8
0.6
0.4
0.2
0.0

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Broad, overlapping components:
coarser readout

Deep, isolated wells:
exact readout

Evidence axis x

b

Entropy

$y_0=2$
$H=0.01$

1 2 3

$y_0=6$
$H=1.18$

5 6 7 8 9

$y_0=11$
$H=2.39$

9 11 13

Readout distribution
over integers

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

c

Predicted RT (s)

2.5
2.0
1.5

Basal latency t_0
Attractor time $\tau E[T]$

Fast, accurate

Slow, approximate

$\mathrm{RT} = t_0 + \tau E[T]$

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Presented numerosity n